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Chapter 31 — Hybrid Mortgage Calculator

A deterministic fixed-rate mortgage calculator with a natural-language interface. The application demonstrates a strict hybrid boundary:

```
User language
|
v
LLM adapter: interpret and explain
|
v
Calculator tool: validate and calculate
|
v
Deterministic financial result
```

The model is never the authority for mortgage arithmetic. It extracts structured inputs and explains the result returned by the calculator; the calculation core owns formulas, validation, precision, and failure semantics.

Learning objectives

Chapter 31 demonstrates how to:

- Separate probabilistic language behavior from deterministic financial computation.
- Implement four inverse fixed-rate mortgage calculations.
- Normalize annual percentages and years into monthly canonical units.
- Validate financial invariants before calculation.
- Use a bounded bisection solver for interest-rate inversion.
- Generate an amortization schedule without premature rounding.
- Expose deterministic behavior through a structured tool contract.
- Use an offline mock adapter for reproducible tests.
- Connect an optional local Ollama model without giving it numerical authority.
- Test both traditional software behavior and natural-language interpretation.

Scope

The MVP supports:

- Fixed-rate mortgages.
- Monthly payment periods.
- Principal (P), monthly periodic rate (r), payment count (n), and monthly payment (M).
- Calculation of exactly one missing primary quantity from the other three.
- Optional amortization schedules of at most 1,200 payment periods.
- Direct structured CLI requests.
- Natural-language requests through the mock adapter or local Ollama.

The MVP does **not** model adjustable-rate or interest-only loans, balloon payments, negative amortization, refinancing, closing costs, mortgage insurance, property taxes, homeowners insurance, HOA fees, lender-specific fees, tax deductions, lender quotes, or personalized financial advice.

Setup

Python 3.12 and uv are required.

```
cd labs/week4/chapter31
uv sync --extra test --extra gui
```

The test environment is self-contained. Tests do not require:

- An API key.
- A running Ollama daemon.
- Network access.
- A visible GUI display; GUI tests use `QT_QPA_PLATFORM=offscreen`.

Quick start

Calculate a monthly payment from a principal, annual interest rate, and term:

```
uv run mortgage calculate \
  --principal 500000 \
  --rate 6.5 \
  --term-years 30
```

Expected text output includes the payment, total paid, total interest, and the scope disclaimer:

```
Principal: $500,000.00
Periodic rate: 0.00541666666666666666666666666667
Annual rate: 6.5000%
Payments: 360
Payment: $3,160.34
...
```

Estimate for principal and interest only; not a lender-specific quote or financial advice.

Run the offline natural-language path:

```
uv run mortgage ask --adapter mock \
  'What is the payment on $500,000 at 6.5% for 30 years?'
```

Run the test suite:

```
uv run pytest -q
```

Direct calculation mode

The `calculate` command accepts exactly three primary quantities:

Input	Meaning	Direct CLI representation
<code>--principal</code>	Loan principal P	Currency amount, e.g. 500000
<code>--rate</code>	Interest rate r	Annual percentage by default, e.g. 6.5; monthly decimal with <code>--rate-period monthly</code>
<code>--payments</code>	Number of monthly payments n	Positive integer, e.g. 360
<code>--term-years</code>	Alias for monthly payment count	Positive term whose value times 12 is an integer, e.g. 30
<code>--payment</code>	Fixed monthly payment M	Currency amount, e.g. 3000

The missing quantity is calculated automatically. `--payments` and `--term-years` are mutually exclusive. Conflicting aliases are usage errors rather than precedence decisions.

Examples:

```
# Calculate payment
```

```
uv run mortgage calculate --principal 500000 --rate 6.5 --term-years 30
```

```
# Calculate principal
```

```
uv run mortgage calculate --payment 3000 --rate 6 --term-years 30
```

```
# Calculate number of payments
```

```
uv run mortgage calculate --principal 500000 --rate 6 --payment 3000
```

```
# Calculate monthly rate, then derive annual rate
```

```
uv run mortgage calculate --principal 500000 --payment 3160.34 --payments 360
```

```
# Supply a monthly decimal rate instead of an annual percentage
```

```
uv run mortgage calculate --principal 500000 --rate 0.005 --rate-period monthly --term-years 30
```

JSON output

Use `--format json` for automation:

```
uv run mortgage calculate \  
  --principal 500000 \  
  --rate 6.5 \  
  --term-years 30 \  
  --format json
```

The JSON response is a discriminated envelope:

```
{  
  "ok": true,  
  "result": {  
    "principal": "500000",  
    "periodic_rate": "0.00541666666666666666666666666667",  
    "payments": 360,  
    "payment": "3160.3401174648186602291583813248289656596159672960",  
    "annual_rate": "0.06500000000000000000000000000000",
```

```

    "term_years": "30",
    "total_paid": "1137722.442287334717682497017",
    "total_interest": "637722.442287334717682497017",
    "missing_quantity": "payment",
    "schedule": null
  },
  "error": null,
  "metadata": {
    "schema_version": "0.2",
    "adapter": "direct",
    "assumptions": [],
    "calculation_config": {
      "solver_tolerance": "1E-12",
      "solver_max_iterations": "100"
    }
  },
  "disclaimer": "Estimate for principal and interest only; not a lender-specific quote or financial adv
}

```

Decimal values are serialized as strings so consumers do not lose precision. Integer counts remain JSON integers. Non-finite values are rejected.

Desktop UI

Launch the PyQt5 desktop application with:

```
uv run mortgage-gui
```

The window has two modes:

- **Calculator:** enter principal, rate, term/payments, and optionally a payment; the UI calculates the missing quantity through the deterministic service.
- **Natural language:** enter a mortgage question and choose **Mock** for the offline adapter or **Ollama** for a local model. Ollama requests run in a worker thread so the UI remains responsive.

The result panel displays payment, principal, annual rate, term, total paid, total interest, assumptions, status/error text, the optional amortization table, and the principal-and-interest disclaimer. Invalid inputs and missing natural-language parameters are rendered as status messages rather than guessed results.

Run the offscreen UI tests with:

```
QT_QPA_PLATFORM=offscreen uv run pytest tests/test_ui.py -q
```

Amortization schedules

Generate a schedule by using the `amortize` command:

```
uv run mortgage amortize \
  --principal 500000 \
  --rate 6.5 \
  --term-years 30 \
  --format json
```

Each row contains:

- period
- payment
- principal
- interest

- balance
- adjusted_payoff

The schedule uses full internal precision. If the ordinary recurrence leaves a residual balance, the final row is an adjusted payoff row whose payment equals principal plus interest and whose balance is zero. Schedule generation is bounded at 1,200 payment periods.

Natural-language interface

Offline mock adapter

The mock adapter is the default and is used by the automated suite:

```
uv run mortgage ask --adapter mock \
  'I can pay $3,000 a month at 6% for 30 years. How much can I borrow?'
```

It supports deterministic examples for:

- Payment calculation.
- Principal calculation.
- Payment-count calculation.
- Rate calculation.
- Clarification when rate or term is missing.
- Unsupported taxes, insurance, HOA, or lender requests.
- Derived principal from purchase price and down payment.
- Payment-too-low validation.

Local Ollama adapter

Install and start Ollama separately, then pull a model:

```
ollama pull llama3.2
```

Run the real adapter:

```
uv run mortgage ask \
  --adapter real \
  --model llama3.2 \
  'What is the payment on $500,000 at 6.5% for 30 years?'
```

Defaults and overrides:

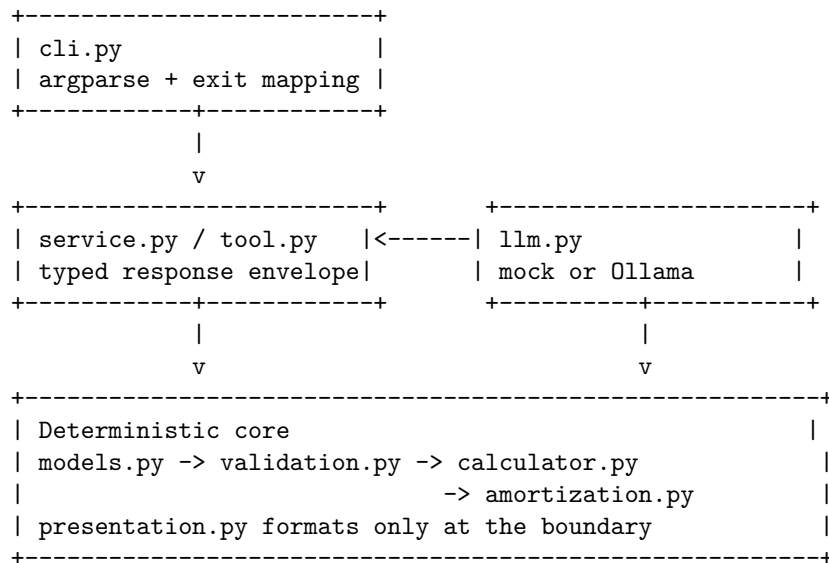
Setting	Default	Override
Ollama host	http://localhost:11434	--host or OLLAMA_HOST
Ollama model	llama3.2	--model or OLLAMA_MODEL
Request timeout	30 seconds	configured by the adapter API

The Ollama adapter makes two non-streaming `/api/chat` requests:

1. Request a JSON-only structured interpretation with normalized monthly fields.
2. Send the calculator result to the model for a natural-language explanation.

The adapter calls the calculator tool between those phases. A timeout, connection failure, malformed model response, or missing `message.content` returns `MODEL_ERROR` with exit code 5. It never substitutes model-generated arithmetic.

Architecture and module map



Module	Responsibility
models.py	Frozen request, result, error, schedule, interpretation, evidence, and metadata types.
validation.py	Quantity-count, domain, finite-number, integer-term, and schedule-bound validation.
calculator.py	Four inverse calculations, zero-rate paths, and bounded bisection rate solving.
amortization.py	O(n) full-precision payment schedule generation.
tool.py	Structured calculator-tool input parsing and Decimal-safe output serialization.
service.py	Public result envelope and conversion of internal failures to typed errors.
llm.py	Offline mock adapter, Ollama HTTP client, and Ollama adapter boundary.
presentation.py	Text/JSON presentation and disclaimer handling.
cli.py	Commands, normalization of CLI aliases, adapter selection, and exit codes.

Financial model

For a positive monthly rate, payment is:

$$M = P * r * (1 + r)^n / ((1 + r)^n - 1)$$

The inverse relationships are:

$$P = M * ((1 + r)^n - 1) / (r * (1 + r)^n)$$

$$n = \ln(M / (M - P*r)) / \ln(1 + r)$$

$$M = P * r * (1 + r)^n / ((1 + r)^n - 1) \quad [\text{solve for } r \text{ by bisection}]$$

At zero rate, the implementation uses:

$$M = P / n$$

$$P = M * n$$

$n = P / M$, only when the quotient is within $1e-9$ of an integer

The rate solver uses the deterministic interval $[0, 1]$, residual/interval tolerance $1e-12$, and at most 100 iterations. A payment that does not cover first-period interest is rejected when solving for the term.

Validation and failure behavior

The public service and tool return an envelope with either **ok: true** and a result or **ok: false** and a structured error. Typical errors include:

Code	Meaning
INVALID_QUANTITY_COUNT	Not exactly one primary quantity is missing.
INVALID_PRINCIPAL	Principal is zero, negative, or non-finite.
INVALID_RATE	Rate is negative or non-finite.
INVALID_PAYMENTS	Payment count is invalid, out of bounds, or a non-integral inverse term.
INVALID_PAYMENT	Payment is zero, negative, or non-finite.
PAYMENT_TOO_LOW	Payment does not exceed first-period interest.
SOLVER_CONVERGENCE	Rate solver cannot bracket or converge on a solution.
CLARIFICATION_REQUIRED	Natural-language input lacks required or unambiguous values.
UNSUPPORTED_SCOPE	Request asks for an excluded financial concept.
TOOL_ERROR	Structured tool input or serialization is invalid.
MODEL_ERROR	Optional real-model request fails or returns malformed output.

CLI exit codes are stable:

Exit	Meaning
0	Successful calculation or clarification response.
2	Usage, normalization, validation, or clarification error.
3	Solver or calculator-tool failure.
4	Unsupported scope.
5	Real-model failure.

Testing

Run all tests with:

```
uv run pytest -q
```

The suite covers:

- Known-value payment calculation.
- Principal, payment-count, and rate inversion.
- Zero-interest behavior.
- Payment-too-low and non-integral-term failures.
- Bisection bracketing and convergence failures.
- Amortization row identity, final payoff, and 1,200-row bound.
- Decimal-string JSON and structured error envelopes.
- Mock natural-language extraction, clarification, scope handling, and evidence.
- Ollama interpretation/explanation sequencing using an offline fake transport.
- CLI output and exit-code behavior.
- Property-based payment/principal and payment/rate round trips over 100 generated cases.

The Ollama adapter tests do not open a socket. To manually exercise the real path, use a locally running Ollama daemon and an installed model.

Development workflow

```
cd labs/week4/chapter31
uv sync --extra test --extra gui
uv run pytest -q
uv run mortgage calculate --principal 500000 --rate 6.5 --term-years 30
uv run mortgage ask --adapter mock 'What is the payment on $500,000 at 6.5% for 30 years?'
```

Keep the deterministic core independent of the model provider. When changing formulas, validation, serialization, or adapter behavior, update the corresponding tests and the authoritative specification at `SPEC.md`.

Safety and scope disclaimer

This project is an educational MVP. Results are estimates for principal and interest only. They are not lender-specific quotes, underwriting decisions, or financial advice. Taxes, insurance, HOA fees, lender fees, and other housing costs are intentionally outside the calculation model.